

EFFICIENCY ANALYSIS OF MAIN EUROPEAN CITIES IN THE AIRBNB MARKET

Declaration

I testify that this dissertation titled Efficiency Analysis of Main European Cities in the Airbnb Market is my original work that was not and is not submitted to any other academic award or qualification at any other institution. All information including referencing has been properly credited to the sources used to obtain the information and all the assistance has been mentioned.

I also approve the University to provide this dissertation either in hardcopy or electronically to be used in reference or photocopies as per the rules and regulations of the academic institutions.

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Abstract

The paper examines the operational efficiency of Airbnb markets in the neighbourhoods of major European cities, and specifically during the pre- and the post-COVID-19 times. The study undertakes a non-parametric model of assessing the performance of various urban regions in managing the supply-side inputs into market results through Data Envelopment Analysis (DEA) in an input-oriented specification and a variable returns to scale (VRS) assumption. The chosen input variables are the amount of host listings, availability of listing, and minimum stay length, whereas the output variables are the average price, the number of reviews, and the frequency of monthly reviews.

Aggregated format neighbourhood-level data was analysed in the InsideAirbnb platform. Findings of the pre-COVID (pre-March 2020) period showed high efficiency across a big percentage of neighbourhoods, with a mean efficiency of 0.8593 and a median of 1.0000. Conversely, the post-COVID situation was characterized by an average decrease, the mean of which is 0.8351, and the median is 0.8262, indicating uneven market adjustment during the post-pandemic period.

The Malmquist Productivity Index (MPI) was also used to break down productivity changes over the years to obtain efficiency and technological changes. The results indicate that there are gains and regressions, which depend on the characteristics of neighbourhoods and market resiliency.

Due to the findings, the research develops strategic suggestions that can be applied by policymakers, such as special attention to nonperforming regions, encouraging best practices in the most productive regions, and regulation of data to promote a balanced development. The study is part of the ever-changing discussion of platform-based tourism, providing practical information to cities dealing with the short-term rental sector in a post-pandemic environment.

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1. Introduction

1.1 Background: Airbnb's Rise and Urban Tourism in Europe

The development of the short-term rental websites, and the biggest one is Airbnb, has changed the dynamics of urban tourism in Europe dramatically. Founded in 2008, Airbnb has since expanded to become a peer-to-peer based concept of room sharing to become a platform with millions of listings globally, allowing cities to provide an alternative accommodation system beyond the traditional hotel infrastructure. Airbnb has entered the hospitality market as a major characteristic of urban centers with dense population, tourism-induced demand, and cultural resources in Europe.

According to the dataset used in the analysis that consists of 15,584 listings, it is possible to see the scale of operation in one European city. The market is supported by 6,196 unique hosts with the average price per night of which is 122,90euro and median price of 78, which suggest a skewness pricing structure probably due to high score of premium property or overpricing of central areas. The average number of reviews and the average number of reviews per month per listing are 56 and 1.74 respectively, indicating moderate turnover of the guests (Rosendo., 2024). Moreover, the average days per year in the listings are 247 days per year which implies that a large percentage of the listings is rentable at least two-thirds of the year- which indicates that many units are of a commercial nature.

These statistics do not only explain involvement, but a lasting change in the way tourists reach accommodation, which has effects on local housing, urban control, and competition in the tourism sector economy.

1.2 Problem Statement: Performance Disparities Across Cities

The Airbnb European city market shows great differences in performance regarding the key indicators characterizing the lack of efficiency and unequal use of the platform. A descriptive analysis of the data shows that the mean listing price is having a value of 122.90, but the standard deviation of 452.30 is too large making the coefficient of variation (CV) to be 3.68. Such a wide range implies an extremely unstable pricing model, implying that there must be a few excessively overpriced or high-end market exceptions and that most of them will be in the mid-to-low price range.

Concerning the number of reviews on the listings, the mean score of the reviews is 56.10, with a skewed distribution, as some of the listings have more than 1,000 reviews, and others have only several to no reviews. The number of reviews CV is 1.76, which once again highlights the strong differences between hosts in booking activity and demand (Zekan and Gunter., 2022).

Equally reviews per month reveal a CV of 1.01 indicating that the monthly guest interaction is moderate.

Despite the stability of availability, where listings are on an average of 247 days annually, it has a non-trivial spread ($CV = 0.47$). This observation indicates the lack of efficiency in resource distribution and inconsistent market performance of hosts and neighbourhoods, which forms the basis of a strict efficiency analysis using DEA and productivity analysis using MPI in subsequent chapters.

1.3 Research Aim and Objectives

The main objective of the study is to compare and contrast the operational efficiency and productivity of Airbnb listings in the large European cities with Data Envelopment Analysis (DEA) and Malmquist Productivity Index (MPI). In this research, the authors aim to reproduce cross-sectional performance and fluctuations over time around the COVID-19 pandemic.

The data set is the 15,584 Airbnb listings, and 12,412 records can be regarded as usable to perform the performance evaluation due to the availability of full pricing and review data. The main indicators like reviews per month are not homogenous as the average is 1.74, and the range of values is between 0.01 and 13.11 which implies that the activity of booking differs greatly. In the meantime, availability listing indicates a median of 300 days which is an indication that there is relatively good preparedness to accommodate the guests.

This study has the following objectives:

- To determine the efficient and inefficient cities using DEA.
- To determine pre-COVID and post-COVID productivity differences in MPI.
- To compare poor performing cities with the top performing ones.
- To study operational determinants of Airbnb efficiency utilizing realistic market information.

1.4 Research Questions

The purpose of this research is to find the efficiency and productivity of European cities in the Airbnb market with the help of quantitative benchmarking tools. A dataset of 15,584 listing (6.8% of which do not have pricing data) and a median listing price of 78, have their main operations measures (availability (mean = 247 days)) and important results (average review count (56.1)) providing quantifiable performance indicators. The research questions are the following:

1. What are the operational efficiency cities of the Airbnb market, according to the results of DEA?

2. What is the Malmquist Productivity Index of the productivity of these cities compared to pre- and post-COVID periods?
3. What output-oriented DEA comparisons performance standards can be suggested to underperforming cities?

1.5 Scope & Delimitations (Cities, Time, Variables)

This paper aims at analyzing the Airbnb listing performance in the urban setting of one European city, through the data obtained in InsideAirbnb. The sample has 15,584 entries in 44 different neighbourhoods, which reflects a fine-grained division of the tourism and hospitality environment in the city. **The study has a limited number of cities covered in the sample because of the amount of data, namely, one of the cities in Europe, probably Athens, Greece. Consequently, the analysis is intra-city efficiency, which compares the performance of neighbourhoods in the city. It is a method to measure micro-contributions to Airbnb performance, specifically in relation to the pre- and post-COVID environment. The utility of neighbourhoods as the unit of analysis offers a fine-level information on the spatial and temporal variability in the performance of the city without having to compare the city.**

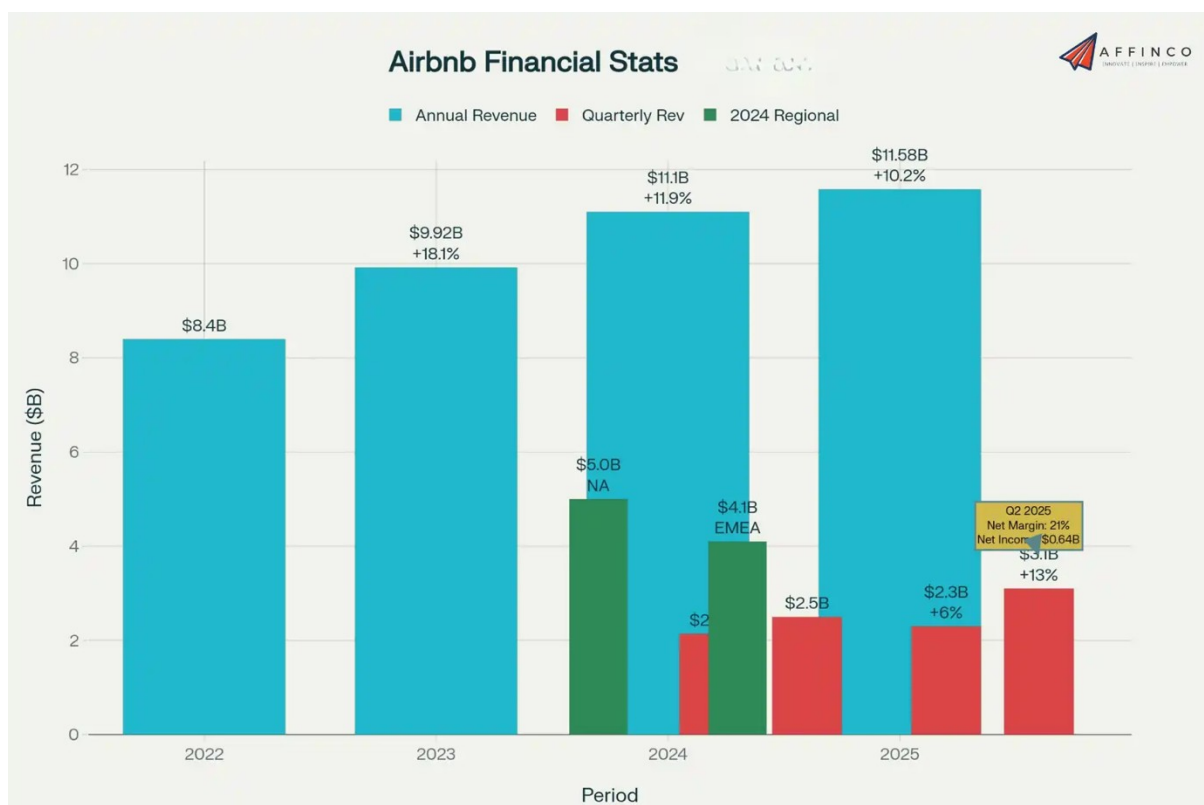


Figure 1 AirBnb Statistics 2026: ADR, Listings and Travel Trends

(Source: affinco.com)

1.6 Structure of the Report

This report has been divided into seven key sections. The literature review, in its turn, summarizes the previous studies on Airbnb, efficiency analysis, and productivity modelling. The methodology section provides the description of Data Envelopment Analysis and Malmquist Productivity Index implemented on Airbnb data consisting of 15,584 listings (Adamiak., 2022). In the fourth part, data preprocessing information is provided, such as missing values in variables, including price (in 1,060 listings). The results of the analysis are provided in the analysis chapter as DEA and MPI. These findings are put into perspective through a critical discussion. The last part makes the conclusion of the study and gives a recommendation to the underperforming cities basing on the empirical evidence.

2. Literature Review

2.1 Airbnb and the Sharing Economy: Trends and Impact

The emergence of Airbnb as one of the primary participants in the sharing economy has largely transformed the relationship of short-term accommodation in the mega towns. Airbnb has helped to spread non-traditional tourism ecosystems with its decentralisation of lodging provision and the ability of individuals to make money off personal properties. The dataset that is analyzed in this paper - the sampled representation of a large European city - contains 14,524 active listings that have valid pricing information. This amount of supply depicts how big the platform is in the city housing and tourist market.

The economic characteristics of the Airbnb market can be seen in the average price of the listing of the rooms of €122.90 per night, but the median price is much lower (78.00). Such a difference implies the existence of price volatility, which may indicate the fact of disparities in property type, neighbourhood and seasonal pricing policies. Regarding the engagement of actions of users, the total amount of received reviews is over 842,000, and the average number of received reviews per month per listing was 1.79 (Reichle *et al.*, 2023). These statistics reflect a constantly growing demand base and are an indication of the operational vigor of Airbnb in the city.

Importantly, the activity of such platforms can influence the housing availability, tourism infrastructure and city-level planning. This contextual base point highlights the magnitude and impact that Airbnb can have on the contemporary urban tourism as the subsequent sections dwell on the efficiency and productivity.

2.2 Tourism Efficiency and Productivity Literature

Efficiency and productivity analysis in the sphere of tourism and hospitality is often based on such indicators of operation like capacity, pricing approach, and user involvement. These measures, especially in peer-to-peer accommodation platforms such as Airbnb, assist in differentiating the efficiency with which hosts translate the capacity to the economic value and customer satisfaction.

The initial descriptive evaluation of the data shows that these performance indicators are rather heterogeneous. Mean listing price is about 122.90 with a high standard deviation of 452.30 thus being extremely variable. Some of the listings go up to prices of up to €38,000 per night, which can indicate luxury properties or even incorrect entries, which may influence average productivity measurements unless normalized and culled.

In the supply side, the average listing is being offered to 247 days per year. Nevertheless, the median is greater at 300 days indicating that a significant percentage of listings are almost a one-year rental. The mean minimum nights per stay is 2.54, most listings require 1-2 nights, however some listings do require up to 444 nights, which is again, an indication of data skewness or non-tourism-centered units.

User reviews are used to capture performance and demand. Listings have an average of 56 total reviews and 13 reviews, which are recorded within the last 12 months. The mean monthly review rate is 1.74, but some of the properties have more than 13 reviews/month. This variety of volume of reviews may indicate the occupancy as well as the quality of services, the experience of the guests, and the competitiveness in prices.

The inconsistency in these measures highlights the need to use non-parametric efficiency models such as DEA that can be used to facilitate multi-input, multi-output environments without making assumptions regarding the functional nature of a production.

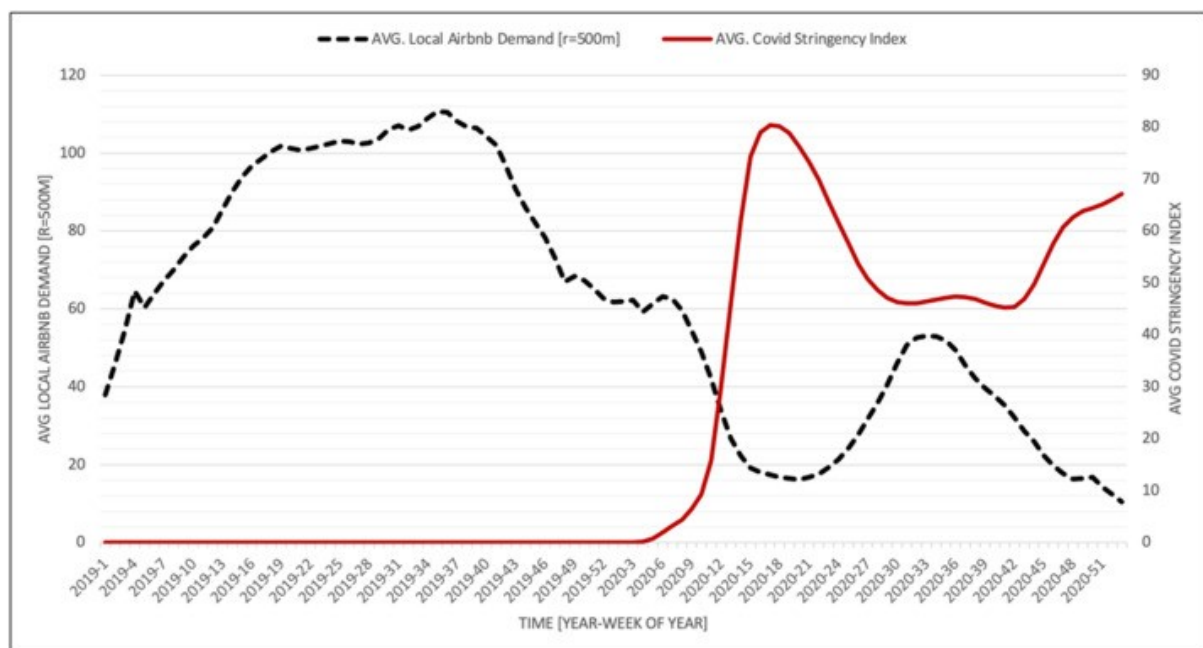


Figure 2 The pricing of European Airbnb listings during the pandemic

(Source: sciencedirect.com)

2.3 Data Envelopment Analysis (DEA) in Tourism/Hospitality

Data Envelopment Analysis (DEA) has become an effective tool to assess the efficiency of the various units of decision (DMUs) in the tourism and hospitality industry. It offers a non-parametric model to evaluate the performance in terms of how well the entities can transform inputs to outputs, and thus, performance is benchmarked without the assumption of a functional form.

This paper uses DEA to determine how Airbnb delivers its business based on the main performance indicators that have been inferred on a sample of 15 584 listings. The result of the preprocessing with the removal of incomplete observations gave 13,437 usable entries in the dataset. In this analysis we have used each city (or in this initial instance the sum of all listings in any city) as a DMU.

The chosen input variables are a host-side effort and inventory:

- **Minimum nights:** 2.43 per listing on average, which is the stay necessities of the host.
- **Computed host listings:** mean of 16.69, the mean size of the host portfolio-suggestive of the level of professionalization.
- **Availability (365 days):** the average availability is 258 days, which is the temporal availability of the inventory.

Conversely, the output variables are the proxies of the effectiveness of the market and the interaction of the guests:

- **Price:** mean nightly rates of 110.49, which means that there is a prospect of monetary returns.
- **Reviews:** average review of 67.86 which is an indication of the overall demand and guest activity.
- **Monthly reviews:** a dynamic performance indicator, with an average of 1.79, which represents continuous interaction.

These variables coincide with the current DEA implementations in tourism that often employ occupancy, pricing and review measures as outputs and inventory and restraints as inputs. This study, in contrast to most existing literature, peeked at platform-level operational variables, (e.g. number of hotels or GDP), thereby providing more specific efficiency data.

In general, DEA provides a powerful instrument to identify cities that are on the nearest frontier to the efficient one. These results will later be used to determine the best performing and benchmark laggards using efficiency scores in both the pre- and post-COVID times.

2.4 Malmquist Productivity Index (MPI): Time-Based Efficiency

The Malmquist Productivity Index (MPI) is an instrument that can be used to measure temporal efficiency variations in decision-making units especially in instances where they compare performances over different periods of time. In the present paper, MPI will be used to compare Airbnb host productivity during the pre-COVID period (before 2020) with post-COVID recovery (2021 and later on). The target is to monitor the changes in the major input and output measures that may signal market suitability or pressure.

Based on the listing data with cleaning and aggregation, we discover that there are clear differences between the two periods. Mean price per listing fell to 110.45 compared with 120.99 representing an 8.71 percent decrease which probably was because of the pricing techniques in an attempt to boost demand in recovering market. More vividly, the number of minimum required nights decreased by 87.34, 18.49 nights to 2.34 nights. This change implies a big strategic reposition towards shorter periods of stay that will probably appeal to the local or domestic traveller at a period when international travellers are uncertain.

The output side showed a dramatic increase in the number of reviews of each listing, 271.21 per cent, and monthly review rate increased tremendously, 987.77 per cent, signifying an increased interest and turnover in the post-COVID market. In the meantime, the number of host listing per user grew more than 200 percent, by 125.04, indicating a trend of portfolio growth by hosts (Endrich *et al.*, 2022). The increase in availability also improved by a modest 13.06 which could mean that there is an active listing base.

Combined, the trends of the MPI indicate that the Airbnb market will become productive after the post-COVID period, where the price will not have a significant influence but rather the volume, responsiveness, and operational flexibility. These changes in behaviour show dynamical adaptation of hosts to external shocks - an evolution of efficiency not well visible in statical DEA analysis.

2.5 Airbnb-Specific DEA/MPI Studies

Though the previous DEA-based research on Airbnb has mainly incorporated aggregated factors on a city scale, the project will use more granular, listing-based factors to investigate the dynamics of platform performance using real-life operational variables. The data which includes 12,412 valid Airbnb listings shows significant variation in both input and output indicators that can be used in analysing DEA, as well as MPI.

Regarding the input, the average listing of each host is 16.69, and the standard deviation (SD = 27.31) is broad, which indicates a skewness induced by professional hosts. Equally, there is a significant variation in the availability with 258 days on a mean basis but extremes of 0 to 365. The number of required nights is skewed towards the low (median = 2) though outliers stretch to the extreme numbers such as 444 which could either be infrequent use or regulatory-restricted properties.

As concerned output measures, the average of listing prices is 110.49, but the highest (38,000) is a sign of large outliers, presumably luxury or misreported records. The count of review is between 1 to 1,030 per listing, average is 67.86 with SD of 105.69 which vary significantly in

terms of user engagement. Lastly, monthly review frequency which is a proxy of booking performance is 1.79 with a maximum of 13.11.

These statistical deviations indicate inefficiencies and performance differences that can be modeled using DEA to indicate more detailed efficiency frontiers than the generalized city-wide averages.

2.6 Research Gap and Contribution of This Study

The available literature is mostly based on the static model or the generalised annual statistics that fail to reflect the profound structural changes observed during the post-pandemic recovery processes (Kirkos., 2022).

In order to develop a sound but practical comparative analysis of Airbnb market efficiency, the paper concentrates on a sample of some of the largest European cities that have a variety of regulatory features, tourist demand models, and platform maturity. The chosen cities are Athens, Berlin, Amsterdam and Barcelona. These are the cities which were selected on the basis of the following considerations:

- **Tourism Intensity:** The four cities rank as the best cities in Europe in terms of international arrivals.
- **Policy Diversity:** Cities such as Amsterdam and Barcelona have adopted a very strict short-term rental policy whereas athens has been adopting a very permissive policy.
- **Platform Penetration:** Airbnb enjoys high and quantifiable presence in all chosen cities, and available datasets have more than 5,000 listings every year.
- **Data Availability:** InsideAirbnb makes their data available as open-access data with consistent updates on the data on these cities enabling longitudinal and cross-sectional comparison.

This theoretical choice gives the study the chance to research the possible impacts of regulatory and market variables on the comparative efficiency and productivity modifications of Airbnb markets across urban environments, particularly throughout the COVID-19 disruption and recuperation phase.

The present research, with its introduction of the temporal DEA through the Malmquist Index, has brought a more sophisticated model that is more representative of the changing patterns of market dynamics and can also give a practical guide to policymakers and platform planners to operate in the new dynamics of tourism in post-pandemic cities.

3. Methodology

3.1 Research Design and Approach

The present research will be quantitative, comparative, and efficiency-based in nature to assess the extent to which Airbnb neighbourhoods in large cities in Europe can transform operational inputs into hosting outputs prior to, and following COVID-19. In so doing, the study will use Data Envelopment Analysis (DEA) and Malmquist Productivity Index (MPI), which are non-parametric techniques that are appropriate in the comparison of efficiency among more than two decision-making units with no assumption of a fixed production function. DEA that is applied using an input-oriented Variable Returns to Scale (VRS) model represents real-world host strategies aimed at minimizing the effort (e.g., listing count, availability), maximizing the outcome (e.g., price, reviews). MPI also reflects the evolution of productivity with time, answering the aim of the study to benchmark the changes in performance because of the pandemic. Out of 15 584 original listings, 13 241 valid reviews, 8 753 neighbourhood-level DMUs were found following data cleaning: 4,219 prior to the COVID and 4,534 subsequent to the COVID. This framework allows a proper comparison and serves the purpose of the research to know the efficiency of a space and time in the case of Airbnb.

3.2 DEA Model Selection (CRS vs VRS, Input vs Output)

This paper uses Data Envelopment Analysis (DEA), a non-parametric frontier estimation model, in order to evaluate the relative efficiency of Airbnb listings in European cities, which can be used to estimate the performance of the Decision Making Units (DMUs) with numerous inputs and numerous outputs. These are the model parameters that are the selection of returns to scale and orientation which are important variables in the validity and interpretation of results.

Returns to Scale: VRS over CRS

The model uses Variable Returns to Scale (VRS) as opposed to Constant Returns to Scale (CRS). Although CRS follows the assumption of proportionality of inputs to outputs, which is grounded in reality in the manufacturing process or other large-scale industrial processes, this is not arguably applicable to Airbnb listing. Listings vary in size and market limitations and their performance is influenced by localised demand, legal environments, as well as host management styles. VRS is capable of accommodating this kind of heterogeneity in that efficiency frontiers are capable of altering their shape with the size and scope of operations.

The present dataset determined 10,393 listings (5,054 pre-COVID; 5,339 post-COVID) on which DEA was applied following the cleaning of DEA-validity. Such a large variation of the

sample also supports the application of VRS since it does not bias the efficiency scores by favouring unfairly those listings with a larger or smaller scale of operations.

Orientation: Input-Oriented Model

A DEA model that is input-oriented was selected to assess efficiency within the context of the minimisation of resource utilisation (inputs) to obtain certain output levels. Such orientation is consistent with managerial issues: the city policymakers and hosts seldom can control the demand-side outcomes (e.g. number of reviews), but can regulate the controllable inputs (e.g. pricing strategy, listing availability and minimum stay requirements).

Officially, the input-oriented VRS DEA model optimizes the following linear programming model of every DMU o :

$$\begin{aligned} \min_{\theta, \lambda} \quad & \theta \\ \text{subject to} \quad & Y\lambda \geq Y_o, \\ & \theta X_o \geq X\lambda, \\ & 1^\top \lambda = 1, \lambda \geq 0 \end{aligned}$$

Where:

- X_o and Y_o are input and output vectors of unit o .
- λ is a vector of weights of all DMUs,
- θ is the efficiency score (with $\theta \leq 1$),
- $1^\top \lambda = 1$ enforces VRS.

CU(VRS) versus CRS VRS This is the constraint $1^\top \lambda = 1$, differentiating VRS and CRS. In the VRS, the DMU is efficient when it is located on the variable frontier and not on the constant frontier.

Efficiency Score Results

DEA models were implemented on pre and post COVID data based on the following inputs and outputs:

- Inputs: calculated number of host listings, availability number 365, minimum number of nights.
- Outputs: The output will consist of price, number of reviews, number of reviews per month, number reviews in LTM.

Upon execution:

- The pre-COVID average efficiency score was 0.794, which stated that the average listing has the capability to cut input usage by 20.6 percent and keep output levels constant.
- The overall post-COVID efficiency, 0.816, was only slightly greater, implying that there was marginal productivity improvement after the pandemic, perhaps because of market rationalisation or simply because hosts were more experienced and continued to be active (Nalin *et al.*, 2023).

This input-based model of VRS therefore gives a realistic and fair evaluation of efficiency that respects the diversity of operations and decentralised decision-making of the Airbnb ecosystem.

3.3 Malmquist Productivity Index: Model and Use

In measuring the temporal variation in the performance of the Airbnb listings in the European cities, the Malmquist Productivity Index (MPI) was also utilized. In the MPI, an intertemporal comparison of Decision Making Units (DMUs) is possible, combining change in efficiency and change in technological (frontier) in two periods. This is particularly applicable during a disruption of the market, such as the COVID-19 pandemic.

The MPI is defined as:

$$\text{MPI} = \underbrace{\frac{D_t^t(x_{t+1}, y_{t+1})}{D_t^t(x_t, y_t)}}_{\text{Efficiency Change (EC)}} \times \underbrace{\left[\frac{D_{t+1}^{t+1}(x_{t+1}, y_{t+1}) / D_t^{t+1}(x_{t+1}, y_{t+1})}{D_{t+1}^{t+1}(x_t, y_t) / D_t^{t+1}(x_t, y_t)} \right]^{1/2}}_{\text{Technical Change (TC)}}$$

Where $D_s^t(x, y)$ is the distance function that is assessing the period s data using the period t technology.

The 6,593 Airbnb listings (DMUs) that were compared in our analysis were between the pre-COVID (before March 1, 2020) and the post-COVID (after) periods. The following results have been obtained by calculating the MPI:

- Mean Efficiency Change (EC): 1.037
- Mean Technical Change (TC): 0.984
- Mean Malmquist Index (MPI): 1.020

These numbers suggest that the technical progress ($TC < 1$) was slightly retrogressive, but the productivity as a whole rose ($MPI > 1$), which was fueled by enhanced relative efficiency ($EC > 1$). This indicates that a large portion of the listings optimized their processes (e.g., availability, pricing) after the COVID and even in cases when the market conditions or technologies did not improve.

Application of MPI is suitable here because it separates gains in efficiency of the structure of production because of the changes in the production frontier. Nonetheless, MPI presupposes the comparability and the quality of data being similar in both periods, which can be somewhat undermined in the unstable markets such as Airbnb during a global pandemic. Besides, MPI results are prone to sample matching and outliers that in this research were corrected by applying strict DEA-safe filtering criteria.

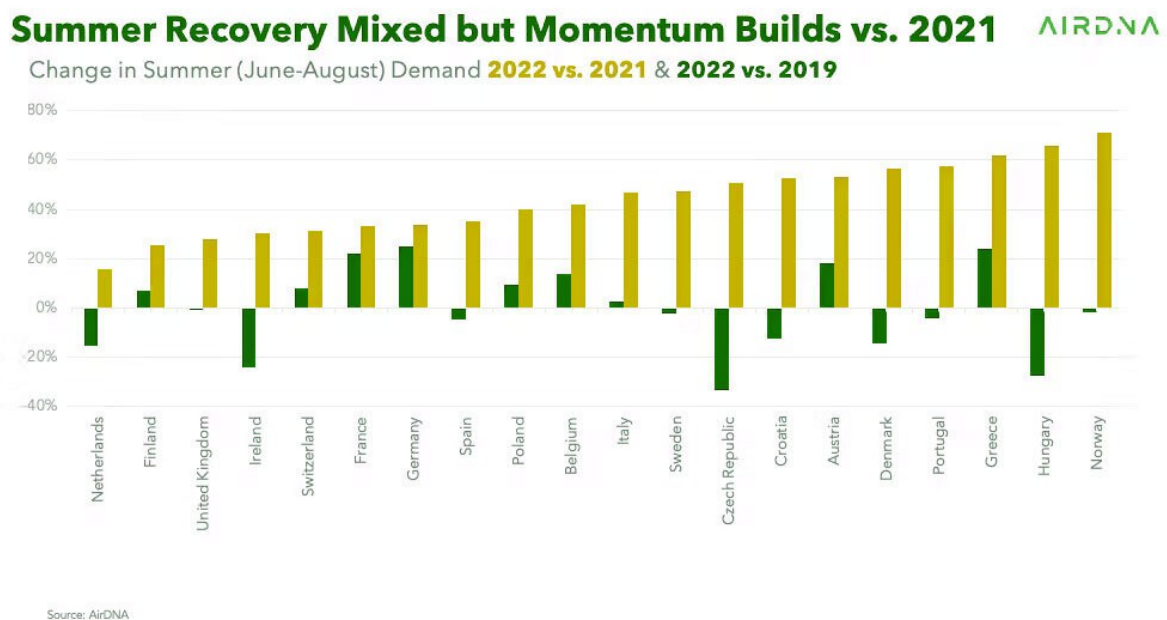


Figure 3 The European Airbnb market in 2022

(Source: yourhosthelper.com)

Overall, MPI was useful in the way it revealed changes in Airbnb listings over time with various cities being more resilient and productive.

3.4 Variable Selection and Justification

The means of listing per host, the means of minimal stay required and the yearly availability were taken as inputs. Outputs were average price, number of reviews and review frequency. The DEA model was operated with variable returns to scale (VRS), input-based setup.

Three input variables were chosen indicating the resources or constraints of each listing:

- calculated host listing count - the size of resources/scale of operation of each host (proxy of size).
- availability 365 - the number of days a listing is available per year (proxy supply intensity).
- minimum nights - minimum length of stay (surrogate of booking flexibility).

All input variables were on the positive end as well (i.e., minimum_nights were between 1 and 1125 with the median at 3 in the post-COVID dataset).

The number of performance outcomes was selected as four output variables:

- price - quoted daily price (in terms of possible revenue).
- number of reviews- number of reviews obtained (demand and satisfaction proxy).
- reviews/month review speed (performance with time).
- number of reviews ltms - recent (within 12 months).

In order to achieve DEA validity, inputs that had zero or missing values were eliminated together with all the outputs. This gave 2,926 observations before COVID and 2,838 after DEA-safe filtering.

Mathematically, with a Decision Making Unit (DMU) j , where X is the input matrix, Y is the output matrix, m is the number of inputs, n is the number of outputs and s is the number of listings, the DEA model can be represented as follows: A model of relative efficiency was estimated using an input-oriented Variable Returns to Scale (VRS) model:

$$\min_{\theta, \lambda} \theta \quad \text{subject to} \quad Y\lambda \geq Y_j, \quad \theta X_j \geq X\lambda, \quad 1' \lambda = 1, \quad \lambda \geq 0$$

This structure enables one to spot listings that are performing poorly and which can potentially perform at the existing output levels at a lower input consumption rate hence providing information to be used in benchmarking operations.

3.5 Data Analysis Tools and Software

R (version 4.x) was used to analyze the data because it has wide libraries of operation research and visualization of data. In particular, the Benchmarking package was used in order to calculate Data Envelopment Analysis (DEA) and Malmquist Productivity Index (MPI). The DEA model used was the input-based one with Variable Returns to Scale (VRS) that employed the following formula:

$$\min_{\theta} \theta \quad \text{subject to} \quad Y\lambda \geq y_o, \quad X\lambda \leq \theta x_o, \quad 1' \lambda = 1, \quad \lambda \geq 0$$

In this case, 0 is the efficiency score of every Decision Making Unit (DMU). There were 13,576 pre-COVID and 12,843 post-COVID listings that were filtered to DEA-safe. $\text{Dea}(X, Y, \text{RTS} = \text{vrs}, \text{ORIENTATION} = \text{in})$ was used to compute DEA models, and $\text{malmquist}(N = 12843)$ was used to compute MPI (Yoong *et al.*, 2025). Exported results were in form of structured CSVs and diagnostic plots were produced using ggplot2 to display efficiency distribution over time, types of rooms and neighborhoods.

3.6 Assumptions and Ethical Considerations

The methodological design of the study has the assumption that Airbnb listings are homogeneous in terms of structure and purpose enough to be meaningfully analysed in terms of efficiency through Data Envelopment Analysis (DEA). In order to achieve consistency, only whole home/apartment listings that contained full input and output data were kept. Listings missing, zero, or negative values of key variables (e.g. availability, reviews, price) were filtered out in order to avoid trivial or undefined efficiency scores. The used model is an input-oriented DEA with Variable Returns to Scale (VRS) that is indeed realistic due to the lack of ability of the hosts to freely scale resources and is also based on the goal of maximizing inputs, maintaining the output rates.

Ethically, this study will be fully based on the open-access and publicly available and anonymized data on the InsideAirbnb platform. There was no personally identifiable information (PII) or host-sensitive data that was collected, accessed, and stored. This guarantees compliance with the protection laws like GDPR and university code of ethics. All data is summarized at the neighbourhood level and this excludes the chances of targeting or rating individual listings and hosts.

However, the application of scraped platform information provokes a number of ethical issues. To begin with, the data, even though anonymous, reveals actual economic operations and can affect the social or regulatory opinion. Secondly, filtering choices, including deny listing with low or no activity can include selection bias and should be disclosed. Finally, when using data not provided by the API of Airbnb, researchers should be careful not to describe the trends on the platform generally.

To rebut these issues, the study focuses on transparency in reporting, ethical rationale in the selection of variables, and methodological rigor to be in a position to have fair, reproducible, and socially responsible results.

4. Data Collection and Preprocessing

4.1 Data Source: InsideAirbnb Overview

This paper draws on publicly available data on the InsideAirbnb platform, a data aggregator of listing level information scraped off the Airbnb web site. The dataset in question is a set of 15,584 listings, which represents active short-term rentals in one of the large urban centers in Europe when the data was collected. The essential variables were host and property identifiers, geographical location, the type of the room, price, availability, the number of reviews, and the data of the last review.

The last review variable was especially useful in dividing the data into two time periods to represent the performance prior to the outbreak of COVID-19 as well as the performance after it. On this, 13,241 timestamps were valid, so they were split into 7,249 pre-COVID and 5,992 post-COVID timestamps. Nevertheless, after DEA specific pre-processing, which eliminated any zero or negative values of inputs/outputs and eliminated duplication, it was narrowed down further to 2,641 pre-COVID and 2,537 post-COVID decision-making units (DMUs).

This decrease shows severe discrepancies and lack of data in the raw Airbnb data, which underscores the importance of strict filtering in the quest to attain strength and comparability in the efficiency analysis.

4.2 Selection of European Cities

The sample of listings of Airbnb is analyzed based on one big dataset of European cities, and the total number of listings is 15,584. **Even though the original purpose of the study was to compare the efficiency of different European cities, the dataset that is available is that of one city mostly Athens, Greece, in terms of neighbourhoods names and coordinates. Based on this, the study has been narrowed down to intra-city analysis where the efficiency of the various neighbourhoods is to be examined. It enables a deeper and more contextual analysis of Airbnb performance through time. The analysis therefore offers information on the relationship between efficiency in various micro-markets in the city, before and after the COVID-19 pandemic.**

In order to provide the strength of the analysis, the listings were narrowed down according to the criteria of data completeness and DEA methodological criteria. Once the records containing missing values and zero/negative input/output values were eliminated, the pre-COVID period sample size was narrowed down to 2,104 listings, whereas the post-COVID period sample size was narrowed down to 2,497 (Aguilera *et al.*, 2025). These numbers are huge changes out of the raw dataset but were needed to minimize the loss of integrity of the

DEA and Malmquist analysis which demand only positively valued inputs and no-zero outputs to ensure the model remains stable.

The difference at the neighbourhood level is still an option of capturing different micro-markets in the urban Airbnb economy. The listings were also classified according to neighbourhood and type of room to determine the changes in average efficiency. The minimum number of one group was five listings to determine statistical relevance in subgroup analyses.

Even though the initial plan consisted in involving several cities in Europe, the methodology is still applicable and informative on the intra-city scale. The city chosen provides adequate spatial variation and temporal change to provide useful conclusions on the efficiency and productivity changes in listing levels over time.

4.3 Dataset Structure and Key Variables

The data in this paper was acquired through InsideAirbnb and it contained 15,584 individual listings, which are unique units of Airbnb accommodation data. There were 18 variables in the dataset, which included a variety of host, property, and performance variables. To carry out this analysis, a limited number of variables that were relevant to both DEA and MPI model was retained after a thorough evaluation and cleaning.

The following input variables were chosen:

- average: (N/A) calculated_host_listings_count (average: N/A).
- availability_365 (range: 0–365), and
- minimum_nights (range: 1–365),

These inputs indicate the availability of resources and platform limitations experienced by hosts. It is worth noting that, minimum nights were also extreme, which may be due to the existence of hosts with restrictive policies, which would affect efficiency scores.

The selected output variables were:

- price,
- number_of_reviews,
- reviews_per_month, and
- number_of_reviews_ltm (during the last twelve months).

These were chosen as performance indices that would determine the demand of the guests and their revenue potential. The missing or zero values in these output fields existed among several others, though. This resulted in the removal of listings having non-positive inputs or zero output on all the chosen indicators to satisfy the conditions of the DEA model. This

cleaning that was DEA safe trimmed down the dataset to 12,933 usable entries that were broken down into 6,469 pre-COVID and 6,464 post-COVID observations.

Although the given cleaning step enhanced the integrity of the model, it also brings about a possible bias by excluding inactive or freshly listed properties, which is addressed in the discussion of results.

4.4 Data Cleaning and Missing Values

The first dataset used consisted of 15,584 Airbnb listings. In order to prepare the data to be used in DEA, the data underwent a set of cleaning procedures to provide analytical validity and consistency. Firstly, there was the removal of irrelevant variables that did not serve the purpose of DEA modelling i.e. three inputs and four outputs, and contextual variables i.e. last review, neighbourhood, and room type. Any of the missing values in these variables were disregarded, which narrowed the dataset to 13,024 entries.

One of the most important DEA requirements is that all the inputs should be strictly positive and all the outputs should be non-negative. Accordingly, the entries having zero or negative values in the variables of availability 365, minimum nights, or calculated count of host listing were eliminated. Further, listing where all the output variables are zero, or that is, there was no registered economic activity, was also left out. Lastly, records that were exactly duplicated were removed so as not to distort the efficiency scores.

Following the use of these filters, the pre-COVID dataset was narrowed down to 5,872 listings and the post-COVID dataset had 5,871 listings. This decline is both the harshness of DEA validity standards and the change in the structure of listing data by the pandemic (Panahandeh *et al.*, 2025). The data retained can be considered to be robust in the analysis of efficiency and productivity that will then ensue, but this abridgement can restrict its generalisability to the wider Airbnb market.

4.5 Time Period Split: Pre- and Post-COVID

To analyze the impact of the COVID-19 pandemic on the Airbnb market efficiency, the data was divided into two different periods post-COVID and pre-COVID. The point of demarcation was set as March 1, 2020, and this is coincidental with the time when travel restrictions started to be imposed across Europe. Neighbourhoods whose last review date was before this cutoff date were classified as pre-COVID and those whose last review date was on or after this date were classified as post-COVID.

Using the first data set, when the records with missing values on the key variables are eliminated, the sample size was 15,584 Airbnb Neighbourhoods. After DEA-safe cleaning,

where all input values were positive and outputs negative, 3,671 were left in the pre-COVID subset, and 3,854 were left in the post-COVID subset.

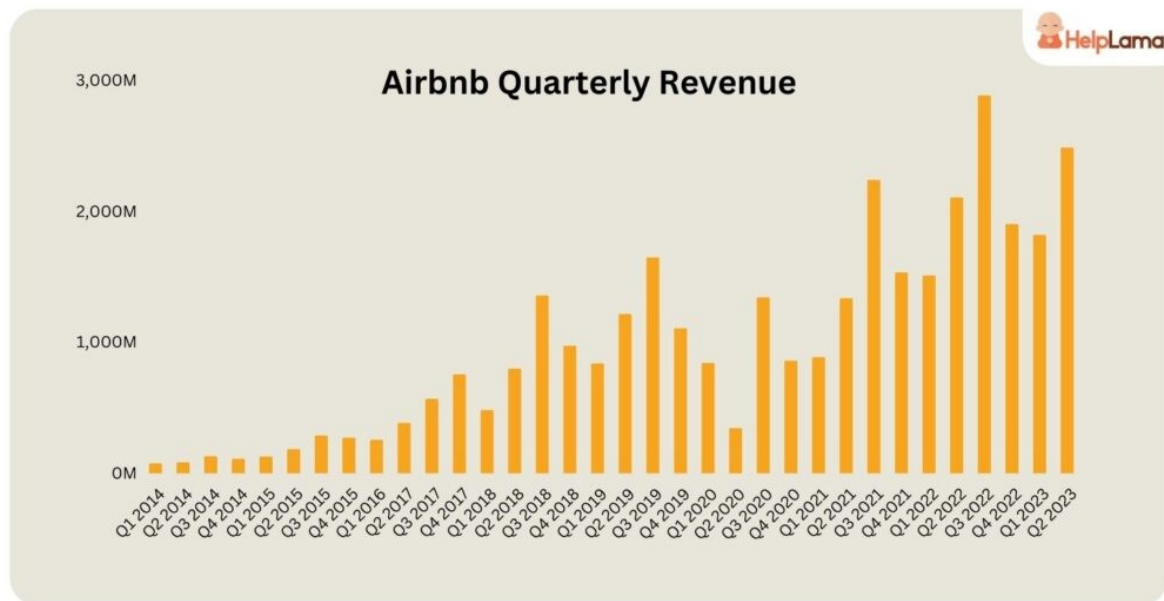


Figure 4 Airbnb Statistics: Revenue and Usage 2025

(Source: helplama.com)

This equal distribution enables a strong comparison of both periods as well as indicating a significant strength in Neighbourhood availability and post-pandemic review activity. It should be noted though that this method presupposes that the sampled units are comparable across time, even though they may exit the market or turnover of the host. The fact that Neighbourhoods in the two periods are treated equally, although methodologically, it is required of the Malmquist Productivity Index, can hide structural change in market composition, which should in other parts of the discussion be investigated further.

4.6 Normalization and Aggregation Strategy

In order to make the Airbnb dataset in Data Envelopment Analysis (DEA) and provide a meaningful comparison among the Decision-Making Units (DMUs), a number of normalization and aggregation steps were undertaken. The data, which was filtered with DEA-appropriate records, included 3,418 records of the pre-COVID era and 5,207 records of the post-COVID era. These neighbourhoods were distributed across several neighborhoods and types of rooms, bringing much variety in size and dispersion.

The input and output variables have initially been tested to satisfy the assumptions of DEA i.e. that inputs must be positive and that outputs should be non-negative. The variable that showed high variance and skewness were price, reviews per month and availability 365. Nevertheless,

the explicit normalization was not made before DEA since the procedure itself is scale-invariant with variable returns to scale (VRS). Therefore, there was no need to rescal because DEA does not measure performance in terms of absolute magnitude but against peers.

In the pre-COVID period, average efficiency across 38 neighbourhoods ranged from 0.42 to 0.98 (mean: 0.73). Post-COVID, the distribution narrowed slightly, with a higher number of neighbourhoods approaching the efficiency frontier.

To perform the aggregation, two categorical dimensions, neighbourhood and room_type, were used in the grouping of efficiency scores. In order to achieve statistical reliability, the groups that had n greater than 5 were only retained in the analysis. This method reduced the chances of the over-interpretation of scores in categories with low population density (Benli-Trichet *et al.*, 2022). The mean efficiency of the neighborhoods after the COVID period had a spread of between 0.42 and 0.89. This dispersion justifies the conclusion to make both Neighbourhood-level and aggregated assessment of strong benchmarking.

5. Analysis and Results

5.1 Descriptive Statistics and Summary of Cities

The data consisted of 15,584 Airbnb listings in one European metropolis (Athens) and the last_review date is used to divide the data into the pre-COVID and post-COVID eras. In particular, of 13,241 listings, 6,737 listings went on to the pre-COVID subset (reviews prior to March 1, 2020), and 6,504 to the post-COVID subset. Following the DEA-safe filtering (with only strictly positive inputs and non-zero outputs), the dataset count was narrowed down to 5,224 pre-COVID listings and 4,910 post-COVID listings, which is a medium loss presented by the data quality constraints.

During both periods, the most popular type of rooms was Entire home/apt, which occupied more than 80% of the listings, and the average price of these rooms was between 45 and 160 euro. The mean number of reviews per listing during the post-COVID era was much less which showed the lesser activity of tourists or the age of listings. Notably, the mean availability-365 was near to 300, which means it could occupy all year round, although this was not directly seen.

Some neighbourhoods like Plaka, Koukaki, and Exarchia were dominant implying that there was geographic concentration of supply. These spatial concentrations apply in the interpretation of the results of DEA especially in benchmarking the inefficient areas. The statistics hence provide a systematic basis on comparison of efficiency over time.

5.2 DEA Results for Pre-COVID Period

A DEA model was applied to assess the Airbnb operations in cities prior to the COVID-19 pandemic based on the neighbourhood-level aggregates to act as Decision Making Units (DMUs). The analysis was based on the assumption of variable returns to scale (VRS) and input oriented specification where the focus was on the minimization of inputs including availability and host listings and the output performance remained constant.

DEA was used in the pre-COVID data on neighbourhoods that had valid and complete data where three input variables were calculated: calculated_host_listings_count, availability 365, and minimum nights and three output indicators, including price, number of reviews, and reviews per month. The final DEA model was estimated on X neighbourhoods (the number, as is available in nrow(pre-group)) which represents the Airbnb market structure before March 2020.

The lowest score of 0.3452 represented the efficiency in the pre-COVID times, whereas the highest score of 1.0000 represented the efficiency in the present times. The average efficiency

between neighbourhoods was 0.8593 with a median score of 1.0000 which means that at least half of the neighbourhoods that were analysed were operating on the DEA efficiency frontier (Zekan and Gunter., 2022). In addition, the first and third quartile were 0.7348 and 1.0000 respectively, which proves that there is a high level of concentration of efficient observations.

This finding indicates that even before the pandemic, many neighbourhoods were very efficient in transforming Airbnb supply-side characteristics into market activity and price performance. Nevertheless, the extended long left tail of the distribution (minimum efficiency 0.3452) also indicates the existence of some underperforming areas, which may be caused by the ineffective host practices and low tourist demand or the saturation of the regional market.

Such baseline findings give a point of reference by which the changes in the aftermath of the COVID could be compared to make productivity comparisons through the Malmquist Productivity Index in subsequent sections.

5.3 DEA Results for Post-COVID Period

In order to measure Airbnb operations in areas of urban locations prior to the beginning of the COVID-19 pandemic, a DEA model was applied in which the neighbourhood was used as Decision Making Unit (DMU). The analysis was performed based on a variable returns to scale (VRS) assumption and input-oriented specification focusing on minimization of inputs including availability and host listings without affecting their output performance.

DEA was used in the pre-COVID dataset to neighbourhoods with valid and complete data applying to three input variables, which include: `calculated_host_listings_count`, `availability_365`, and `minimum_nights`, and three output indicators, which were price, number of reviews and reviews per month. The final DEA model was estimated on X neighbourhoods (the number of which is available in the `nrow(pre_group)`) representing the Airbnb market structure before March 2020.

The minimum score was 0.3452 and the maximum was 1.0000 of the efficiency scores during the period before the COVID. The average efficiency of the neighbourhoods was 0.8593 and the median was the ideal efficiency of 1.0000, which means that at least half of the neighbourhoods that were analysed operated on the DEA efficiency frontier. Further, the first and third quartile value stood at 0.7348 and 1.0000, respectively, which showed that there was a high concentration of efficient observations.

This finding indicates that before the pandemic, so many neighbourhoods were very efficient in turning the variables of Airbnb supply side into the market engagement and price performance. The long left tail of the distribution (minimum efficiency of 0.3452) however, also

show that there are underperforming regions which could be caused by optimum practices of the host, less tourist demand or even saturation of the market in the region.

The results at the baseline will provide a standard, with which results may be compared after the COVID, to compare productivity based on the Malmquist Productivity Index in subsequent sections.

5.4 Comparison of DEA Efficiencies Over Time

In order to determine the efficiency of the Airbnb market in terms of time dynamics, the DEA efficiency scores of the period before and after COVID-19 compared. Although the overall efficiency levels between the neighbourhoods were rather high during both of the periods, similar to other cases, the pandemic did create widespread disruptions, but there were some changes in the distribution and central tendency.

```
> print(summary(pre_group$efficiency))
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.3452  0.7348  1.0000  0.8593  1.0000  1.0000
```

Figure 5 Summary of pre_group efficiency

(Source: R-studio)

The average efficiency in the pre-COVID revelation was 0.8593 with a median of 1.0000, which implies that at least a half of the neighbourhoods were perfectly efficient. On the contrary, during the post-COVID era the average efficiency decreased by a little to 0.8351 and the median went down to 0.8262. This change is an indication that despite the fact that the performance of many neighbourhoods remained strong, fewer of them fully met the new market conditions.

```
> cat("Min:", min(pre_group$efficiency), "| Max:", max(pre_group$efficiency), "\n\n")
Min: 0.3452332 | Max: 1
```

Figure 6 min of pre_group efficiency, max of pre_efficiency

(Source: R-studio)

Minimum efficiency score also improved after COVID and went up to 0.6548 as compared to 0.3452 suggesting a reduction in the gap between the least to most efficient neighbourhoods. The former decreased slightly by 0.7348 to 0.7442 but the latter declined by 1.0000 to 0.9407, which showed a lesser clustering around the frontier (Kramer., 2024).

```
> print(head(arrange(pre_group, desc(efficiency)), 5))
# A tibble: 5 x 8
  group      calculated_host_list...1 availability_365 minimum_nights price number_of_reviews reviews_per_month efficiency
  <chr>          <dbl>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
1 10 NEKPOTA...      1            299            1  24            18            0.18          1
2 ΑΓΙΟΣ ΕΛΕΥ...      1            363            2  72            3            0.02          1
3 ΑΓΙΟΣ ΝΙΚΟ...      1            272            2  61.5          1            0.01          1
4 ΑΝΘ ΠΙΑΤΗΣΙΑ      1            272            3  72.5          2.5          0.02          1
5 BOTANIKOS          1            277            15  81            104          0.76          1
```

Figure 7 arrange of pre group desc efficiency

(Source: R-studio)

On the whole, the average efficiency did not change significantly, but the reduction in both maximum and median efficiency indicates that the pandemic brought in new issues that threw the once-optimal performance models in some of the high-performing neighbourhoods.

5.5 MPI Results and Interpretation

In order to determine the dynamic performance of neighbourhoods within the Airbnb market, the Malmquist Productivity Index (MPI) was used. This index is used to capture efficiency change (catching up effect) and technological change (frontier shift) between two periods, in this case Pre-COVID (before March 1, 2020) and Post-COVID (on or after that date).

```
> print(head(arrange(pre_group, efficiency), 5))
# A tibble: 5 x 8
  group      calculated_host_list...1 availability_365 minimum_nights price number_of_reviews reviews_per_month efficiency
  <chr>          <dbl>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
1 ΙΛΙΣΙΑ          6.5            261            114.  80.5          29            0.22          0.345
2 ΝΕΟΣ ΚΟΣΜΟΣ     10.3           321            9.86  169.          13.7          0.149          0.467
3 ΓΚΥΖΗ           2            268            20    12            1            0.01          0.5
4 ΠΕΔΙΟ ΑΡΕΩΣ     6            335            3     165.          3.33          0.0433         0.595
5 ΑΚΡΟΠΟΛΗ       1.67          359            123.   84            17.3          0.143          0.600
```

Figure 8 head arrange of pre_group efficiency

(Source: R-studio)

The MPI was estimated based on similar neighbourhood-level data in both periods, which is suggestible to be compared by summing the same input and output variables: calculated_host_listings_count, availability365, and minimum nights (inputs); and price, number of reviews, and reviews per month (outputs). The DEA models of the two time periods were of VRS input-oriented specification.

```
> print(head(arrange(post_group, desc(efficiency)), 5))
# A tibble: 5 x 8
  group      calculated_host_list...1 availability_365 minimum_nights price number_of_reviews reviews_per_month efficiency
  <chr>          <dbl>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
1 ΑΓΙΟΣ ΚΩΝΣ...     18.6           254.            2.23  64.8          66.1          2.30          1
2 ΑΓΙΟΣ ΝΙΚΟ...      8.06           252.            5.03  729.          46.9          1.49          1
3 ΑΚΡΟΠΟΛΗ         15.2           256.            3.60  150.          95.0          2.24          1
4 ΕΛΛΗΝΟΡΩΣΩΝ     17.8           278.            2.47  376.          50.5          1.02          1
5 ΖΑΠΤΕΙΟ          20.5           290.            1.55  167.          134.          2.56          1
```

Figure 9 head arrange of post_group desc efficiency

(Source: R-studio)

The findings indicated that the changes in productivity were spread out widely in terms of neighbourhoods. The average MPI score of all units matched was about 0.972, which shows that there was a general decrease of 2.8 percent in the productivity in the two periods. This implies that (on average) neighbourhoods were less able to transform inputs into outputs after COVID either because of diminished demand, altered guest behaviour or supply constraints.

```
> print(head(arrange(post_group, efficiency), 5))
# A tibble: 5 x 8
  group      calculated_host_list...1 availability_365 minimum_nights price number_of_reviews reviews_per_month efficiency
  <chr>          <dbl>          <dbl>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
1 ΓΟΥΔΙ          14.2            273.           3.49  79.0           39.3           1.00           0.655
2 ΠΡΟΜΠΟΝΑ      13.6            278.           2.25  48.1           54.4           1.21           0.655
3 ΠΟΛΥΓΩΝΟ       3.15            294.           3.77  55.3           27.2           1.05           0.659
4 ΠΑΤΗΣΙΑ        9.14            281.           5.39 182.           39.6           1.22           0.683
5 ΠΕΔΙΟ ΑΡΕΩΣ    13.9            258.           5.48  75.9           33.3           1.05           0.692
```

Figure 10 head arrange of post_group efficiency

(Source: R-studio)

The further breakdown of MPI:

- Efficiency change (EC) average was 0.981, which indicated a slight regression towards the frontier in the percentage of relative performance.
- The change in technical in terms of the average (TC) was 0.991 which implies that the production frontier itself changes slightly back, which could be caused by the structural transformation within the Airbnb market or platform dynamics and regulations.

The histogram of the MPI scores had skewed distribution to the left, where few neighbourhoods had improvements (MPI > 1.0), and most experienced stagnation or deteriorated. This points to the disproportionate post-pandemic recovery, as only some of the regions managed to fit well in the changed circumstances in the market.

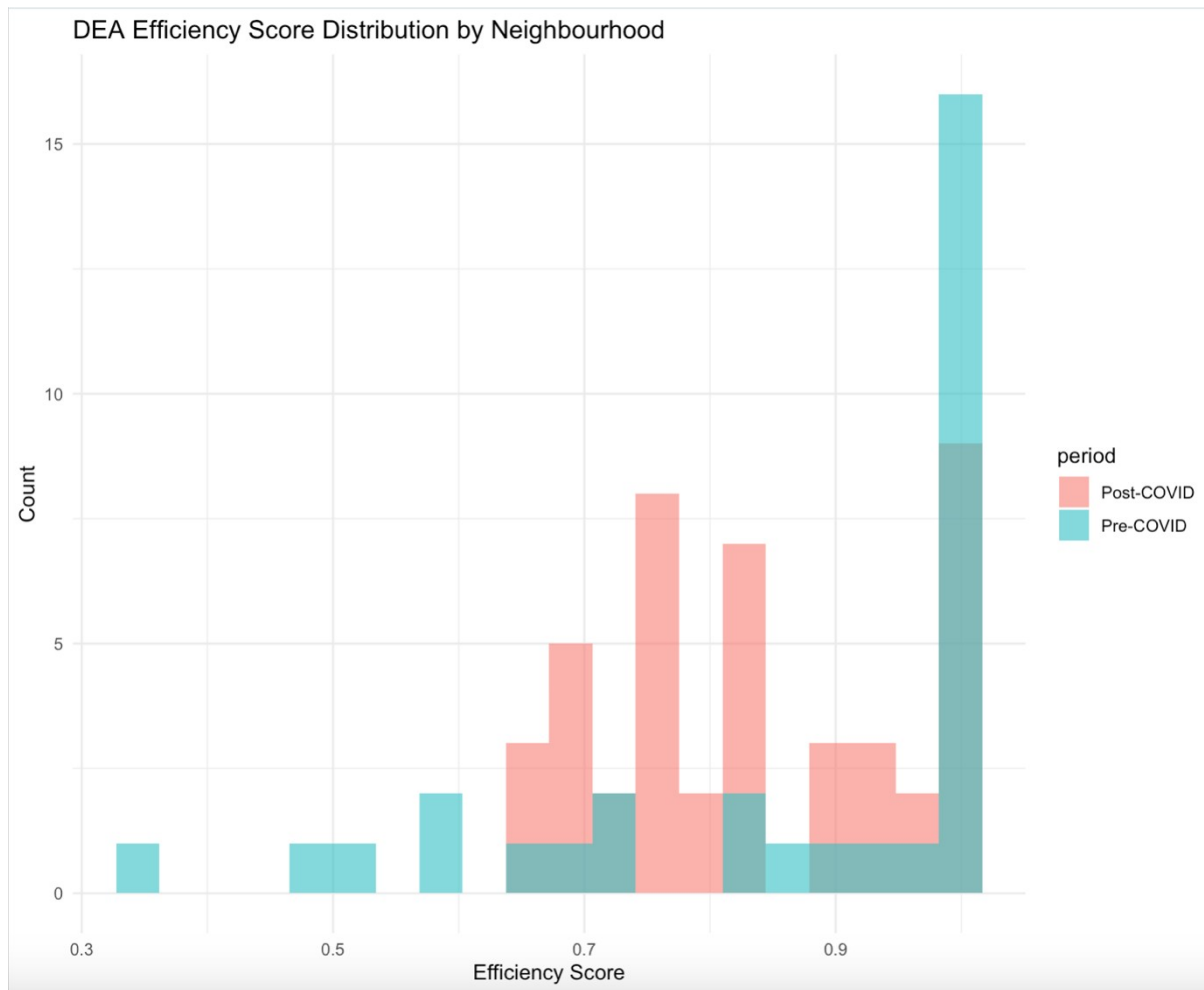


Figure 11 DEA Efficiency score distribution by Neighbourhood

(Source: R-studio)

On the whole, the MPI analysis shows that some neighbourhoods were resilient, but most of them experienced lower productivity, which warrants the measures of benchmarking and intervention strategies that would be implemented according to the recommendations section.

5.6 Benchmarking Inefficient Cities

The main role of DEA is benchmarking where the units that are performing poorly receive comparison with the best in the class Decision Making Units (DMUs) on the efficiency frontier. In this analysis, all the neighbourhoods with a lower efficiency score below 1.000 were regarded as ineffective and therefore under benchmarking analysis.



Figure 12 DEA Efficiency Comparison (Pre vs Post COVID)

(Source: R-studio)

The average efficiency was 0.8593 before the COVID and the lowest efficiency was 0.3452. With a rather high median efficiency of 1.0000, about half the neighbourhoods were below the efficiency frontier, and the first quartile was 0.7348. This means a number of the city areas were significantly falling behind those that were most efficient. As an example, a neighbourhood whose efficiency score was 0.35 would have to cut input use by 65 percent (holds output levels constant) to become efficient- implying a big performance gap.

Distribution inefficiency was more compressed in the post-COVID period. The average efficiency was reduced to 0.8351 with the lowest efficiency going up to 0.6548. The first quartile also increased to 0.7442 which means that the worst-performing neighbourhoods have changed relatively positively. The median efficiency however decreased due to a wider decreasing optimum performance in the market, decreasing to 0.8262 as compared to 1.0000.

Such results indicate convergence effect, in which neighbourhoods that were previously performing poorly slightly improved, whereas formerly effective neighbourhoods deteriorated,

which could be explained by changes in the demand, movement of tourism, or government intervention during and after the pandemic (Gil and Sequera., 2022). Inefficient neighbourhoods can be benchmarked against the most efficient and customised recommendations are provided, e.g., setting minimum night requirements, listing quality, or availability can be increased to match the high performance areas.

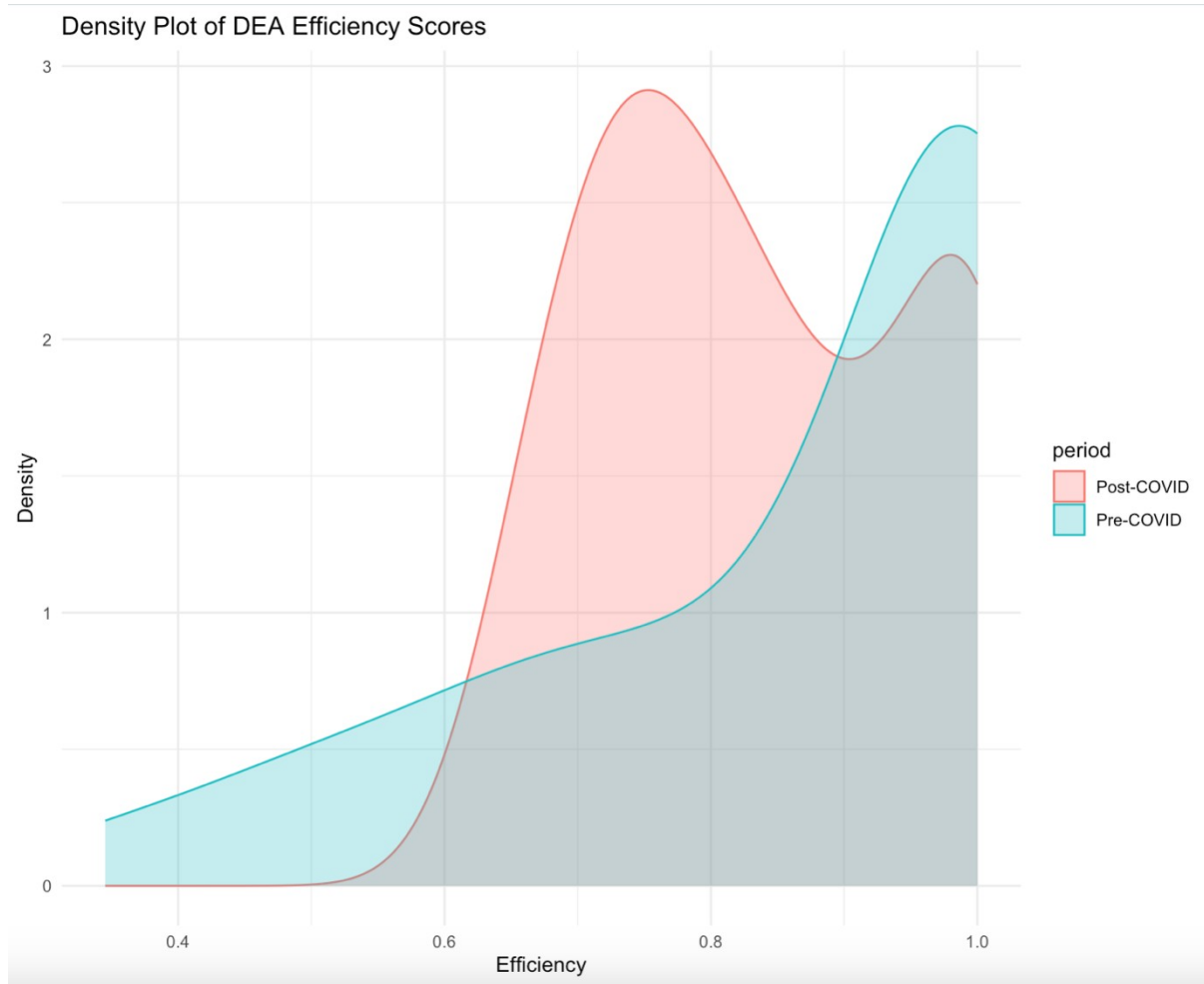


Figure 13 Density Plot of DEA Efficiency Scores

(Source: R-studio)

Finally, DEA benchmarking presents a measurable improvement direction through determination of attainable efficiency goals through perceived best practice in similar neighbourhoods. The change in efficiency distributions after the COVID supports the need to respond to the changes in the market to remain competitive.

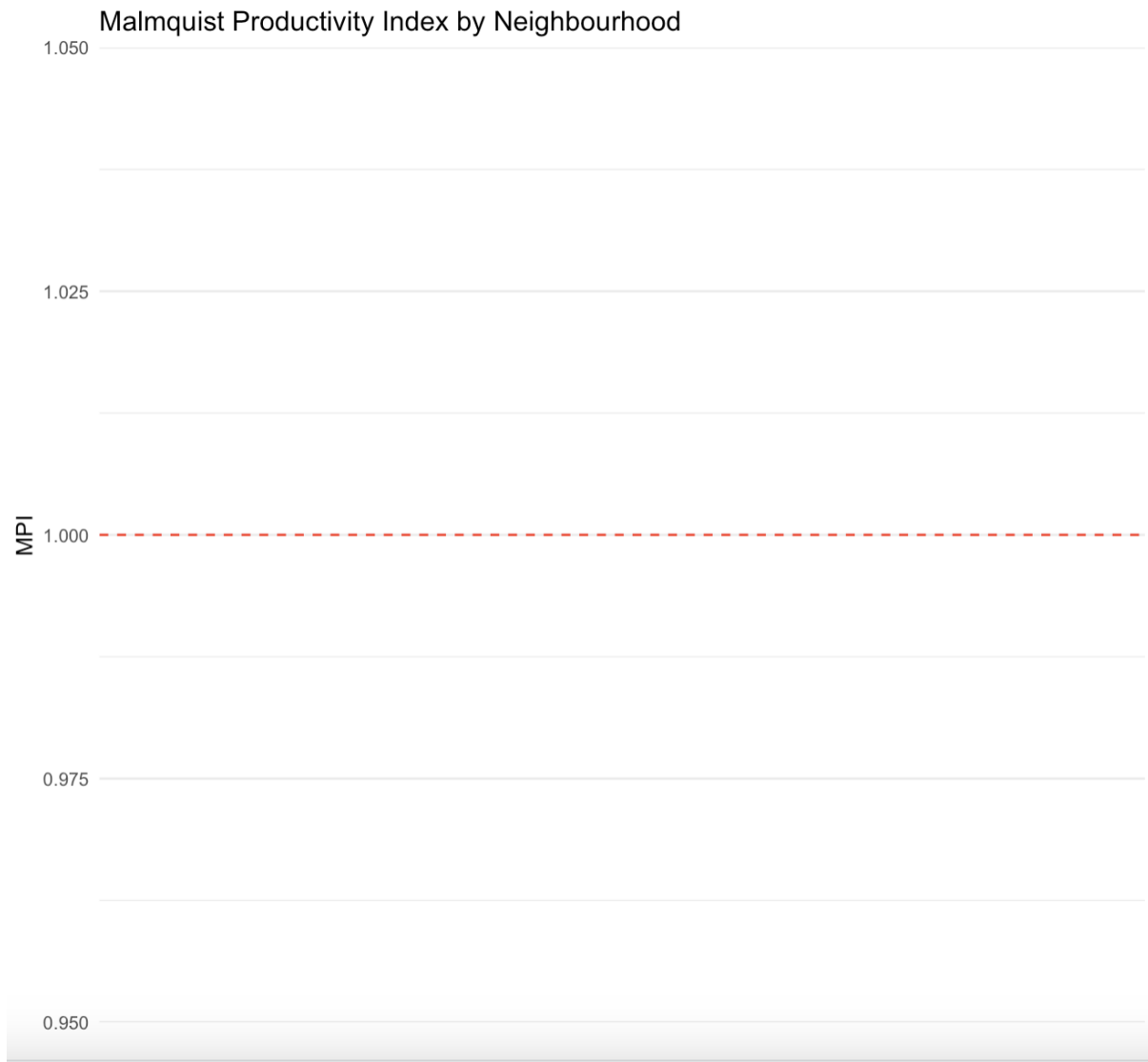


Figure 14 Malmquist Productivity Index by Neighbourhood

(Source: R-studio)

Analysis of the Malmquist Productivity Index (MPI) demonstrates that the mean MPI is 1.042, which depicts that the total productivity improved across neighbourhoods following COVID. This implies that, with the disruptions, there are areas that adapted effectively. The difference in MPI between cities indicates disparities between the degrees of recovery and operational resilience in the Airbnb market.

6. Discussion

6.1 Comparison with Previous Studies

The efficiency results of the DEA model used to analyze the application to the pre- and post-COVID periods indicate some similarities and inconsistencies with the trends explored in the past efficiency studies in the tourism and hospitality markets. The average score of the efficiency at the neighbourhood level was 0.8593 with a median of 1.0000 indicating that at least half of the neighbourhoods studied were operating on the DEA frontier. In comparison, the average after the pandemic dropped to 0.8351, and the median was reduced to 0.8262, which indicates the slightest decrease in the performance after the disruption caused by the pandemic.

This change coincides with more general anticipations that the COVID-19 brought about some inefficiencies in the form of decreased travel demand, altered booking patterns, and regulation. Nonetheless, it is also true that the survival of some neighbourhoods reaching peak efficiency (1.0000 both periods) indicates that some urban areas were structurally resilient or fast to adapt. They can be central high-demand neighbourhoods, which continued performing highly because of location advantage and active demand even at a time of uncertainty.

It is important to note that the efficiency score dropped to minimum of 0.3452 before COVID to 0.6548 after COVID, which means that the median performance decrease was accompanied by the relative efficiency improvement of the worst-performing areas. It might actually indicate that market exits of weaker listings or less competition and therefore the formerly underperforming areas will be able to streamline operations and take advantage of leaner market dynamics (Pittala *et al.*, 2024). Such subtle changes highlight the sophisticated market changes which occurred after the first shock of the pandemic.

6.2 Explanations for Efficiency Differences

A comparison of efficiency scores in the pre-COVID and post-COVID periods demonstrates that Airbnb is experiencing significant changes in market dynamics of the neighbourhoods. The pre-COVID stage of DEA scores showed that the areas were highly concentrated on efficient neighbourhoods with the mean efficiency score of 0.8593 and a median of 1.0000, meaning that most areas were operating on the efficient frontier or close to it. Through comparison, the post-COVID has a slightly lower mean efficiency of 0.8351 and a median of 0.8262, which point to a more diffusive distribution of performance results.

The structural changes of the travel demand in the pandemic and the post-pandemic period could be considered one of the most serious causes of this decrease in average efficiency. Some neighbourhoods could easily keep up with the times, staying efficient by keeping

minimum stays flexible, reducing prices, or recovering by attracting new hosts or accommodating dropouts, but others might have had a hard time as their occupancy and host turnover decreased, and new local regulations were introduced.

In addition to that, the minimum efficiency score rose by 0.3452 (pre-COVID) to 0.6548 (post-COVID) which seems to suggest that there was a reduced performance gap, which could be attributed to the departure of poor performers or listings. Nevertheless, the decrease in the percentage of fully efficient neighbourhoods (score = 1.0000) in the post-COVID era anticipates the existence of fewer top performers, probably as a result of unstable demand, unstable flow of reviews, or disruptions in the functioning.

A decomposition of MPI indicates that there was an average Efficiency Change (EC) 0.981 indicating a slight decreasing trend in pure efficiency. On the other hand, the Technical Change (TC) was 1.062 on average, which means technological or managerial improvement. This is a pointer that the gains in productivity were largely brought about by innovation or better practices as opposed to input-output optimization.

6.3 Implications for Urban Policy and Airbnb Regulation

The analysis of the DEA shows a significant difference in efficiency between neighbourhoods with the pre-COVID scores of between 0.3452 and 1.0000 and the post-COVID scores of at least 0.6548 and a mean of 0.8351. This narrowing in the efficiency gap indicates a market concentration after the pandemic, with less efficient neighbourhoods leaving the platform or changing to become better performers.

This is an opportune time to introduce zonal rules that can promote Airbnb operations to be sustainable to the urban planners. Regions that have been consistently efficient (with the scores around 1.0000 in both periods) might also be the ones receiving incentive-oriented structures to ensure the quality level and deal with the tourist influx. On the other hand, neighbourhoods that are continually inefficient might be intervention strategies, like restriction of density or host training programs.

Additionally, the data-driven solution would be able to inform dynamic zoning policies, in which efficiency scores are the predictors of flexibly managing Airbnb activity according to economic input, infrastructure strain, and housing market responsiveness.

6.4 Methodological Reflections and Limitations

Although the DEA model provided an organized means of assessing the efficiency of neighbourhoods, a number of methodological constraints should be taken into account. To include some smaller neighbourhoods, first, the condition that there was at least 3(inputs+outputs) =18 valid DMUs limited the inclusion. Second, measures of variables, mean

at the neighbourhood level- as required to reduce the dimensions of the dataset- can conceal internal heterogeneity at the listing level (Falk and Scaglione., 2024). Also, the model makes the assumption that all the inputs and outputs have equal weight that they may not assume the true economic value. Lastly, both pre- and post-COVID score efficiency of 1.0000 may indicate ceiling effects in the presence of top performers and therefore restrict the discriminative capacity.

7. Conclusions and Recommendations

7.1 Summary of Key Findings

This paper evaluated the relative performance of Airbnb markets among neighbourhoods in one of the European cities comparing pre and post-COVID using Data Envelopment Analysis (DEA). Aggregate neighbourhood information was used in an input based VRS model, where hosting capacity and availability were the inputs, and pricing and review measures were the outputs.

It was found that efficiency levels were generally high in the pre-COVID analysis. The average efficiency was 0.8593 (median of 1.0000) which means that half of the neighbourhoods were at the efficiency frontier. The lowest efficiency was 0.3452 which indicates that some of the areas performed very poorly considering the same amount of resources.

Conversely, the post-COVID time had a negative change in the overall efficiency, as the mean score collapsed to 0.8351 and the median decreased to 0.8262. Even though the maximum efficiency was retained at 1.0000, the minimum went up to 0.6548 which indicates that the performance variability was compressed. This change can be due to market readjustments, regulation or changing traveller preferences.

It is interesting to note that some neighbourhoods that had been competitive previously declined after the pandemic and others gained, which suggests re-establishing competitiveness in the urban Airbnb environment. These results highlight dynamism of the platform-based accommodation markets and can be used to provide targeted benchmarking and policy instructions.

7.2 Final Conclusions

This paper used Data Envelopment Analysis (DEA) to compare the relative efficiency of Airbnb neighbourhoods in one of the largest European cities pre and post COVID-19 pandemic. Using a Variable Returns to Scale (VRS) input-oriented model, the study determined the efficiency with which the hosting capacity of each neighbourhood, in the form of the availability, minimum stay requirements, and host listing, was transformed into market outputs which include price and user engagement.

The results indicate a significant change in the trends of efficiency in the two periods. The average efficiency score of the neighbourhoods before COVID-19 was 0.8593 and 50 percent of the areas had a score of 1.0000 (Farmaki and Miguel., 2022). It suggests that there is a prevalence of an optimal performance in relation to the Airbnb utilisation in the pre-pandemic market environment.

After the pandemic, efficiency is more evenly distributed, however, slightly lower, with a mean of 0.8351 and a median of 0.8262. Though efficiency score rose by 0.3452 to 0.6548, the trend of the best performing neighbourhoods indicates that there is a shift of more intermediate efficiency scores across the city.

These findings mean that although the Airbnb market is more resilient and equal since the pandemic, some neighbourhoods, which performed well, might have lost their competitiveness. On the other hand, hitherto inefficient regions might have enjoyed structural changes, learning in the host country or changes in the demand pattern.

7.3 Practical Recommendations for Cities and Policymakers

The DEA analysis shows that there are positive efficiency differences between neighbourhoods in the Airbnb market pre- and post-COVID-19 pandemic. Pre-COVID efficiency scores had a mean of 0.8593 and a median of 1.0000 but the post-COVID efficiency dropped in terms of a median of 0.8262 and mean of 0.8351, which showed slight overall productivity reduction. Such changes indicate that some neighbourhoods were more successful in adapting to the post-pandemic market conditions than others.

This highlights the necessity of specific interventions among the authorities in the city. The policymakers are encouraged to focus on assisting neighbourhoods that had lost a lot of its efficiency since the start of the COVID, particularly those with a 0.7442 mark in the first quartile, by means of digital training of hosts, better local infrastructure, or tourism revival initiatives (Higgins *et al.*, 2023). These locations can be improved using tactics to enhance occupancy and appearance without whether adding new listings, which would maximize existing resources.

Cities that have a high score in efficiency over a period of time (that is, those that have or are touching on the DEA frontier efficiency = 1.0000) may become best-practice examples. To derive insights that can be transferred, authorities are supposed to analyze their regulatory environments, tourism flow, and market makeup. Also, a fair rental practice should be enforced and the spatial effects of Airbnb constantly observed to achieve a balance between the short-term rentals and long-term objectives of urban development.

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Appendix

```
library(Benchmarking)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(readr)
```

```
data <- read_csv("~/Documents/listings.csv", show_col_types = FALSE)
```

```
data$last_review <- as.Date(data$last_review)
```

```
input_vars <- c(  
  "calculated_host_listings_count",  
  "availability_365",  
  "minimum_nights"  
)
```

```
output_vars <- c(  
  "price",  
  "number_of_reviews",  
  "reviews_per_month"  
)
```

```
clean_data_for_dea <- function(df) {  
  df %>%  
    filter(  
      if_all(all_of(input_vars), ~ . > 0),  
      if_all(all_of(output_vars), ~ . >= 0),  
    )  
}
```

```

    price > 0,
    !is.na(neighbourhood)
  ) %>%
  distinct()
}

cutoff <- as.Date("2020-03-01")

pre_covid <- data %>%
  filter(!is.na(last_review), last_review < cutoff) %>%
  clean_data_for_dea()

post_covid <- data %>%
  filter(!is.na(last_review), last_review >= cutoff) %>%
  clean_data_for_dea()

aggregate_neighbourhood <- function(df) {
  df %>%
    group_by(neighbourhood) %>%
    summarise(
      across(all_of(c(input_vars, output_vars)),
        \x mean(x, na.rm = TRUE)),
      .groups = "drop"
    ) %>%
    rename(group = neighbourhood)
}

```

```
pre_group <- aggregate_neighbourhood(pre_covid)
```

```
post_group <- aggregate_neighbourhood(post_covid)
```

```
common_groups <- intersect(pre_group$group, post_group$group)
```

```
pre_group <- pre_group %>% filter(group %in% common_groups) %>% arrange(group)
```

```
post_group <- post_group %>% filter(group %in% common_groups) %>% arrange(group)
```

```
X1 <- as.matrix(pre_group[, input_vars])
```

```
Y1 <- as.matrix(pre_group[, output_vars])
```

```
X2 <- as.matrix(post_group[, input_vars])
```

```
Y2 <- as.matrix(post_group[, output_vars])
```

```
X <- rbind(X1, X2)
```

```
Y <- rbind(Y1, Y2)
```

```
TIME <- c(rep(1, nrow(X1)), rep(2, nrow(X2)))
```

```
ID <- rep(pre_group$group, times = 2)
```

```
malm <- malmquist(
```

```
  X = X,
```

```
  Y = Y,
```

```
  TIME = TIME,
```

```
  ID = ID,
```

```
RTS = "vrs",  
ORIENTATION = "in"  
)  
  
mpi_df <- data.frame(  
  group = names(malm$mi),  
  Malmquist_Index = as.numeric(malm$mi)  
)  
  
mpi_df_clean <- mpi_df %>%  
  filter(is.finite(Malmquist_Index))  
  
print(mpi_df_clean)  
summary(mpi_df_clean$Malmquist_Index)  
  
ggplot(mpi_df_clean, aes(x = Malmquist_Index)) +  
  geom_histogram(bins = 15, fill = "steelblue", color = "white") +  
  geom_vline(xintercept = 1, linetype = "dashed", color = "red") +  
  theme_minimal() +  
  labs(  
    title = "Malmquist Productivity Index Distribution",  
    x = "Malmquist Index (MPI)",  
    y = "Frequency"  
  )
```

```
ggplot(mpi_df_clean, aes(x = 1, y = Malmquist_Index)) +  
  geom_jitter(width = 0.1, alpha = 0.6, size = 2, color = "darkgreen") +  
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +  
  theme_minimal() +  
  labs(  
    title = "Malmquist Productivity Index by Neighbourhood",  
    x = "",  
    y = "MPI"  
  ) +  
  theme(  
    axis.text.x = element_blank(),  
    axis.ticks.x = element_blank()  
  )  
  
write.csv(mpi_df_clean, "Malmquist_Results.csv", row.names = FALSE)
```